Chemical disinfectants are grouped by the types of microbes and infectious agents they are effective against.

High-level germicides kill vegetative cells, fungi, viruses and endospores and can ultimately lead to sterilization.

Intermediate-level germicides cannot kill all viruses and are less effective against endospores.

Low-level germicides kill vegetative cells and some enveloped viruses, but are ineffective against endospores.

3.4 Major Groups of Antimicrobial Chemical Agents

A large number of chemical agents are in common use. Some of the more common groups are listed below.

1. Phenol and Phenolics

Phenol was the first widely used chemical antiseptic and disinfectant. In 1867, Joseph Lister employed carbolic spray to reduce the risk of infection in surgical theatres. Phenol derivatives called phenolics contain altered molecules of phenol useful as antiseptics and disinfectants. The phenolics damage cell membranes and inactivate enzymes of microorganisms, while denaturing the proteins. Phenolics includes cresols, such as Lysol, as well as several bisphenols, such as hexachlorophene. Today phenol and phenolics such as cresol, xylenol, and orthophenyl phenol are used as disinfectants in laboratories and hospitals.

The commercial disinfectant Lysol is made of mixture of phenolics. Phenolics are tuberculocidal, effective in the presence of organic material, and remain active on surfaces long after application. However, they have a disagreeable odour and can cause skin irritation.

Hexachlorophene is one of the most popular antiseptics because it persists on the skin once applied and reduces skin bacteria for a long period. However, it can cause brain damage and is now used in hospital nurseries to control and outbreak of gram positive (*Staphylococcal*) infections. It is mainly used in soaps and creams. It is an ingredient of various dermatological preparation used for skin disorders.

2. Alcohols

Alcohols are among the most widely used disinfectant and antiseptic. They are bactericidal and fungicidal but not sporicidal. Alcohols can destroy the lipid component of enveloped viruses. The two most popular alcoholic germicides are ethanol and isopropanol. They act by denaturing proteins and dissolving membrane lipids. The recommended optimum concentration of ethanol is 70%, but concentration between 60% and 95% are employed to kill germs as well. Thermometers and small instruments are disinfected by soaking in alcohol for 10 to 20 minutes.

3. Halogens

Halogen compounds are broad spectrum compounds that are considered low toxicity, low cost and easy to use. Among the halogens, iodine and chlorine are important antimicrobial agents. Small quantities of drinking water can be disinfected with halazone tablets.

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